

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET E — CASE STUDY & SOURCE-BASED SCENARIOS****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-E-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET E - Case Study & Source-Based Scenarios):

1. The question paper comprises **5 In-Depth Case Studies & Source Scenarios** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each source passage, historical inscription, geographic case, and economic scenario carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: SYNOPTIC MECHANICS OF THE INDIAN SOUTHWEST MONSOON

The Indian Meteorological Department (IMD) monitors several interconnected macro-scale atmospheric and oceanic systems to forecast the arrival and intensity of the Southwest Monsoon. During late spring, intense solar insolation over the vast continental landmass of Central and Northwest India creates an intense thermal low-pressure cell. Simultaneously, the Intertropical Convergence Zone (ITCZ)—the equatorial trough of low pressure—migrates northward to position itself over the Indo-Gangetic plain. At the same time, the elevated Tibetan Plateau acts as an immense high-altitude heat engine, heating the middle troposphere and generating a strong upper-level anticyclone that unleashes the Tropical Easterly Jet stream around latitude 14°N. In the Southern Indian Ocean, east of Madagascar, a powerful high-pressure system known as the Mascarene High intensifies. Driven by the pressure gradient between the Mascarene High and the North Indian thermal low, southeast trade winds cross the equator, get deflected to their right by the Coriolis force, and barrel into the subcontinent as moisture-saturated Southwest Monsoon winds.

Q.1 What planetary force deflects southeast trade winds into southwesterly winds as they cross the equator toward India?

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|--|--|
| (A) Gravitational attraction of the moon | (B) The Coriolis force induced by the Earth's rotation |
| (C) Magnetic pull of the North Pole | (D) Centrifugal force of deep ocean tides |

Q.2 What role does the seasonal northward migration of the ITCZ play in the onset of the Indian monsoon?

- | | |
|--|--|
| (A) It brings freezing polar blizzards to the Deccan Plateau | (B) It establishes an intense low-pressure trough over the northern plains, drawing maritime moisture inward |
| (C) It stops all clouds from forming across South Asia | (D) It causes the Indian Ocean to freeze solid |

Q.3 Why does the elevated Tibetan Plateau function as an atmospheric thermal engine during summer?

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|---|---|
| (A) It is covered in active molten lava flows year-round | (B) Its high elevation absorbs intense solar insolation, heating the mid-troposphere and pumping convective air aloft |
| (C) It reflects 100% of solar radiation back into outer space | (D) It is situated directly on the geographic equator |

Q.4 What is the Mascarene High, and how does it influence Indian monsoon rainfall?

- | | |
|--|---|
| (A) A high-pressure cell east of Madagascar that acts as the primary southern source region pumping winds toward India | (B) A tropical cyclone that permanently batters Sri Lanka |
| (C) A deep ocean trench near Indonesia that swallows seawater | (D) A mountain peak in the Western Ghats of Kerala |

Q.5 What upper-tropospheric jet stream develops over peninsular India during the summer monsoon season?

- | | |
|--|--|
| (A) The Polar Night Westerly Jet | (B) The Tropical Easterly Jet stream (around 14°N) |
| (C) The Subtropical Westerly Jet stream (south of Himalayas) | (D) The Gulf Stream oceanic current |

CASE STUDY 2: TROPICAL CYCLOGENESIS & SUPERCYCLONE FANI OVER THE BAY OF BENGAL

In May 2019, Extremely Severe Cyclonic Storm Fani struck the coast of Odisha near Puri, packing sustained winds of over 215 km/h. Tropical cyclogenesis requires specific thermodynamic and kinematic atmospheric conditions: a warm ocean sea surface temperature (SST) exceeding 26.5°C to 27°C, high middle-tropospheric relative humidity, significant Coriolis force to initiate spin (cyclones cannot form within 5° of the equator), and low vertical wind shear between the surface and upper troposphere. Over the warm waters of the Bay of Bengal, rapid evaporation supplied vast quantities of water vapour. As moist air ascended into an existing low-pressure disturbance, water vapour condensed into towering cumulonimbus clouds, releasing immense amounts of latent heat of condensation. This latent heat warmed the core, lowering central surface pressure further to 932 hPa and accelerating spiraling gale winds. At the center of Fani lay a classic cloud-free, calm 'eye' of subsiding air, surrounded by a terrifying 15-kilometre-wide 'eyewall' of maximum winds and torrential rains.

Q.6 What is the primary thermodynamic fuel that powers and intensifies a tropical cyclone over ocean waters?

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|--|--|
| (A) Friction against continental mountain ranges | (B) Latent heat of condensation released when ascending moist maritime air condenses into clouds |
| (C) Nuclear fission inside deep ocean trenches | (D) Cold ocean currents chilling the lower troposphere |

Q.7 Why do tropical cyclones never form directly on the geographic equator (between 0° and 5° latitude)?

- | | |
|--|---|
| (A) The ocean temperature at the equator is below 0°C | (B) The Coriolis force is virtually zero at the equator, preventing the initiation of cyclonic rotation |
| (C) The equator experiences zero atmospheric pressure year-round | (D) Trade winds are completely absent at the equator |

Q.8 What meteorological conditions prevail inside the 'eye' of a mature tropical cyclone?

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|---|--|
| (A) Torrential downpours and the highest wind speeds of the storm | (B) Calm winds, subsiding air, lowest atmospheric pressure, and relatively clear skies |
| (C) Freezing snow blizzards and giant hail storms | (D) Massive volcanic ash clouds blocking all sunlight |

Q.9 Where are the most destructive, catastrophic winds and heaviest rainfall concentrated within a tropical cyclone?

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|--|---|
| (A) In the outer rain bands 500 km away from the storm | (B) In the eyewall immediately surrounding the central calm eye |
| (C) Directly in the center of the eye | (D) Behind the cold front after the storm dissipates |

Q.10 Why do tropical cyclones rapidly weaken and dissipate soon after making landfall on the continental mainland?

- (A) They lose their primary moisture source (warm ocean water) and encounter increased surface friction
- (B) The sun shines brighter over continental landmasses
- (C) City buildings shoot lasers to disperse cyclone clouds
- (D) The Coriolis force immediately disappears over dry land

CASE STUDY 3: OROGRAPHIC PRECIPITATION EXTREMES: MAWSYNRAM VS SHILLONG RAIN SHADOW

The state of Meghalaya exhibits one of the most dramatic spatial rainfall contrasts on planet Earth. Mawsynram, situated on the southern crest of the Khasi Hills at an altitude of 1,400 metres, receives an astonishing average annual rainfall of approximately 11,872 mm, holding the global record for the wettest inhabited place. In stark contrast, the state capital Shillong, located only 55 kilometres to the northeast on the northern leeward slopes, receives barely 2,200 mm per year. The geomorphic alignment of the Garo, Khasi, and Jaintia Hills forms a funnel-like trap open to the Bay of Bengal. When moisture-laden Southwest Monsoon winds strike this barrier, they are forced to ascend abruptly. As the saturated air ascends the windward slopes, it expands and cools at the saturated adiabatic lapse rate (approximately 5°C to 6°C per 1,000 m). Heavy condensation triggers relentless orographic downpours. Once the air crests the ridge and descends the northern leeward slope toward Shillong, it compresses adiabatically, warms up, its relative humidity plummets, and cloud formation is suppressed, creating a classic rain shadow.

Q.11 What primary topographic mechanism produces the extraordinary precipitation at Mawsynram?

- (A) Cyclonic frontal collisions between polar air and desert air
- (B) Orographic lifting of moisture-bearing Bay of Bengal winds funneled by the Khasi Hills
- (C) Artificial cloud seeding conducted by local tea plantations
- (D) Continuous thermal convection from geothermal hot springs

Q.12 Why does Shillong receive less than one-fifth of the annual rainfall recorded at nearby Mawsynram?

- (A) Shillong is located in a hot hyper-arid desert plain
- (B) Shillong lies on the leeward side of the hills in a dry rain-shadow zone where descending air warms and suppresses condensation
- (C) Shillong is located south of the equator
- (D) Monsoon winds never reach the state of Meghalaya

Q.13 What happens to the temperature and relative humidity of an air parcel as it is forced to ascend a mountain barrier?

- (A) Temperature rises and relative humidity drops to zero
- (B) Temperature drops due to adiabatic expansion, and relative humidity rises toward 100% (saturation)
- (C) Temperature and humidity remain completely constant
- (D) The air turns into pure liquid water immediately at ground level

Q.14 Which mountain ranges form the funnel-shaped topographic trap in Meghalaya?

- (A) Aravalli, Vindhya, and Satpura Ranges
- (B) Garo, Khasi, and Jaintia Hills
- (C) Zaskar, Ladakh, and Karakoram Ranges
- (D) Nilgiri, Anaimalai, and Cardamom Hills

Q.15 Which another region of India demonstrates a classic rain-shadow effect during the Southwest Monsoon?

- (A) The Malabar Coast of Kerala
- (B) The Western Ghats' western slopes
- (C) The Deccan Plateau (e.g., Marathwada/Vidarbha) on the eastern leeward side of the Western Ghats
- (D) The Sunderbans delta of West Bengal

CASE STUDY 4: URBAN HEAT ISLAND (UHI) EFFECT AND WINTER THERMAL INVERSION IN DELHI

During November and December, the National Capital Region (NCR) of Delhi experiences severe public health crises characterized by thick, toxic smog. Environmental scientists trace this seasonal disaster to a combination of anthropogenic emissions and seasonal meteorological dynamics: the Urban Heat Island (UHI) effect coupled with radiation thermal inversion. Under normal daytime tropospheric conditions, temperature decreases with height (normal environmental lapse rate), allowing warm polluted surface air to rise and disperse into the upper atmosphere. However, during calm, clear winter nights, the dry land surface radiates longwave terrestrial infrared heat into space with extreme rapidity. The air layer directly in contact with the ground cools much faster than the air aloft. This sets up a 'temperature inversion', where a warm air layer sits atop a cold, dense surface air layer. The warm inversion lid completely halts vertical convective air mixing, trapping vehicular particulate matter ($PM_{2.5}$ and PM_{10}), industrial smoke, and agricultural stubble burning emissions in a stagnant, toxic ground-level smog.

Q.16 What is a 'temperature inversion' (thermal inversion) in atmospheric science?

- | | |
|---|---|
| (A) When temperature drops faster than 20°C per kilometer | (B) When a layer of warm air overlies a colder air layer near the ground, reversing the normal lapse rate |
| (C) When boiling hot lava erupts from subterranean aquifers | (D) When absolute zero temperature is reached at sea level |

Q.17 How does a winter thermal inversion lid contribute to severe air pollution episodes in North India?

- | | |
|---|--|
| (A) It causes hurricane-force winds that blow all smog into the ocean | (B) It halts vertical convective atmospheric mixing, trapping smoke, exhaust, and particulate matter close to the ground |
| (C) It converts carbon dioxide into pure breathable oxygen | (D) It increases solar ultraviolet rays that dissolve all smoke particles |

Q.18 What causes the rapid ground cooling on clear winter nights that triggers radiation inversion?

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|---|---|
| (A) Rapid loss of terrestrial heat via unobstructed longwave infrared radiation into space under clear, dry skies | (B) Direct absorption of ice crystals falling from the stratosphere |
| (C) The Earth temporarily stops rotating for several hours | (D) Heavy thick cloud blankets reflecting sunlight |

Q.19 What is the 'Urban Heat Island' (UHI) effect observed in large metropolitan cities?

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|---|--|
| (A) Cities being physically located on volcanic oceanic islands | (B) Urban areas experiencing significantly higher temperatures than surrounding rural areas due to concrete, asphalt, and anthropogenic heat |
| (C) Cities being colder than snow-covered mountain peaks | (D) Urban centers having zero atmospheric air pressure |

Q.20 What is the toxic winter mixture of smoke, dust, and condensed moisture droplets called?

- | | |
|------------------------|------------------|
| (A) Smog (Smoke + Fog) | (B) Cirrus cloud |
| (C) Auroral borealis | (D) Glacial till |

CASE STUDY 5: THE ANTARCTIC OZONE HOLE & THE MONTREAL PROTOCOL GLOBAL TREATY

In 1985, atmospheric scientists from the British Antarctic Survey published shocking satellite and balloon data revealing a catastrophic seasonal depletion of stratospheric ozone (SO_3) over Antarctica, colloquially named the 'Ozone Hole'. The culprit was identified as synthetic halocarbons, specifically chlorofluorocarbons (CFCs) and halons widely used in refrigeration, air conditioning, and aerosol spray cans. Under ordinary tropospheric conditions, CFCs are chemically inert

and non-toxic. However, over decades, they drift upward into the stratosphere. During the freezing Antarctic winter night, temperatures plunge below -78°C , forming Polar Stratospheric Clouds (PSCs) composed of nitric acid trihydrate and water ice. On the crystal surfaces of PSCs, inactive chlorine reservoirs (such as HCl and ClONO_2) react to liberate reactive chlorine molecules (Cl_2). When spring sunlight returns in September, ultraviolet radiation photo-dissociates Cl_2 into free chlorine radicals ($\text{Cl}\cdot$). A single chlorine radical catalytically destroys up to 100,000 ozone molecules: $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$; $\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$. In 1987, the global community signed the Montreal Protocol, phasing out CFCs in one of the most successful environmental treaties in human history.

Q.21 In which atmospheric layer is the protective ozone layer located, and what is its primary biological function?

- (A) Troposphere; producing thunderstorm rain showers (B) Stratosphere; absorbing lethal solar ultraviolet (UV-B and UV-C) radiation to protect living organisms
- (C) Mesosphere; reflecting commercial FM radio waves (D) Exosphere; generating polar auroral light displays

Q.22 Why does ozone depletion occur most severely over Antarctica during the early southern spring (September–October)?

- (A) Antarctica has the highest concentration of factories producing aerosol cans (B) Extreme winter cold forms Polar Stratospheric Clouds (PSCs), whose crystal surfaces catalyze the release of destructive chlorine radicals upon spring sunrise
- (C) The South Pole is located closest to the Sun in the month of September (D) Volcanoes in Antarctica emit 100% of global carbon dioxide

Q.23 How does a single chlorine radical ($\text{Cl}\cdot$) destroy thousands of stratospheric ozone molecules?

- (A) It acts as a catalyst that destroys ozone while being continuously regenerated in a self-sustaining catalytic cycle (B) It physically freezes the ozone gas into solid blocks of ice
- (C) It absorbs all oxygen atoms from the entire planet (D) It transforms ozone into radioactive uranium isotopes

Q.24 Which international environmental agreement successfully phased out the production and consumption of ozone-depleting substances?

- (A) The Kyoto Protocol (B) The Montreal Protocol (1987)
- (C) The Treaty of Versailles (D) The Geneva Convention

Q.25 What major health consequence would result from unmitigated depletion of the Earth's stratospheric ozone layer?

- (A) Widespread blindness from cataracts, skin carcinomas, and suppression of phytoplankton photosynthesis in marine food chains (B) Immediate freezing of the global human blood supply
- (C) Immunity to all infectious viral diseases (D) A permanent drop in global sea levels

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & CASE STUDY ANALYTICAL EXPLANATIONS (SET E)

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.1	(B)	According to Ferrel's Law, the Coriolis force deflects air moving north of the equator to its right, turning southeast trades into southwest monsoons.
Q.2	(B)	The northward shift of the ITCZ over the Gangetic plain creates the 'monsoon trough' that attracts maritime air from the southern oceans.
Q.3	(B)	The high elevation of the Tibetan plateau acts as an elevated heat source, warming the mid-troposphere and driving upper-level easterly jet streams.
Q.4	(A)	The Mascarene High is a semi-permanent subtropical high-pressure cell near Madagascar whose strength directly regulates the intensity of cross-equatorial monsoon flow.
Q.5	(B)	The Tropical Easterly Jet stream steers monsoon depressions into the Gangetic valley and sustains vigorous convective precipitation.
Q.6	(B)	Latent heat released during the condensation of water vapour warms the storm core, driving pressure lower and feeding the cyclonic engine.
Q.7	(B)	The Coriolis force is proportional to $\sin \phi$; at the equator ($\phi = 0^\circ$), Coriolis force is zero, making cyclonic vortex spin impossible.
Q.8	(B)	The eye is a central zone of dry subsiding air characterized by light winds, clear skies, and the lowest barometric pressure of the cyclone.
Q.9	(B)	The eyewall is a dense ring of towering cumulonimbus clouds surrounding the eye containing the storm's most violent winds and heaviest rain.
Q.10	(A)	Landfall cuts off the cyclone from warm ocean moisture (latent heat engine) while high topographic ground friction decelerates wind speeds.
Q.11	(B)	Orographic rainfall occurs when moisture-laden winds are forced to rise abruptly over mountain barriers, cooling and dropping massive rain.
Q.12	(B)	Air descending the leeward slope warms adiabatically; its moisture capacity rises and relative humidity drops, creating a rain shadow.
Q.13	(B)	Ascending air expands under lower atmospheric pressure, cooling adiabatically. Cooling brings temperature to the dew point, achieving saturation.
Q.14	(B)	The Garo, Khasi, and Jaintia hills of the Meghalaya plateau funnel southwesterly Bay of Bengal monsoon winds into a tight dead-end.
Q.15	(C)	The Western Ghats force incoming Arabian Sea winds to drop heavy rain on the western coast, leaving the interior Deccan Plateau in a semi-arid rain shadow.
Q.16	(B)	Thermal inversion is a reversal of normal tropospheric lapse rate, where temperature increases with altitude, trapping cold dense air below.
Q.17	(B)	The warm inversion layer acts as an impermeable ceiling, preventing buoyant vertical dispersion and trapping pollutants at human breathing height.
Q.18	(A)	Clear skies and low humidity allow unhindered loss of longwave terrestrial radiation into space, chilling the ground and surface air.
Q.19	(B)	Urban Heat Islands develop because concrete, asphalt, vehicular engines, and reduced vegetation absorb and re-emit far more heat than rural countryside.

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.20	(A)	Smog is a hazardous atmospheric mixture of smoke particles, chemical pollutants, and condensed fog droplets.
Q.21	(B)	The stratospheric ozone layer filters harmful high-energy ultraviolet radiation, shielding terrestrial life from genetic damage and skin cancers.
Q.22	(B)	Heterogeneous chlorine activation requires Polar Stratospheric Cloud ice surfaces formed during polar night, triggering rapid catalytic ozone loss at spring sunrise.
Q.23	(A)	Chlorine radicals operate as catalysts: $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$ and $\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$, continuously regenerating Cl to destroy up to 100,000 O_3 molecules.
Q.24	(B)	The Montreal Protocol (1987) is the landmark international environmental treaty mandating the global phase-out of CFCs and halons.
Q.25	(A)	Unfiltered solar UV-B radiation damages DNA, causing severe melanoma skin cancers, cataracts, and marine ecological collapse.